SELECTED WORK · 2023–2026

Charles Abi Chahine.

architect · computational designer

design, computation, and the work of getting it built.

https://charlesabichahine.com

CONTENTS

Eight of eighteen. The rest at https://charlesabichahine.com/work

Sensi.

MACAD 2026 AWARD

A sensory copilot for floor plans: it reads a plan and scores how each room will feel across six coupled senses, personalized to the person who will live in it.

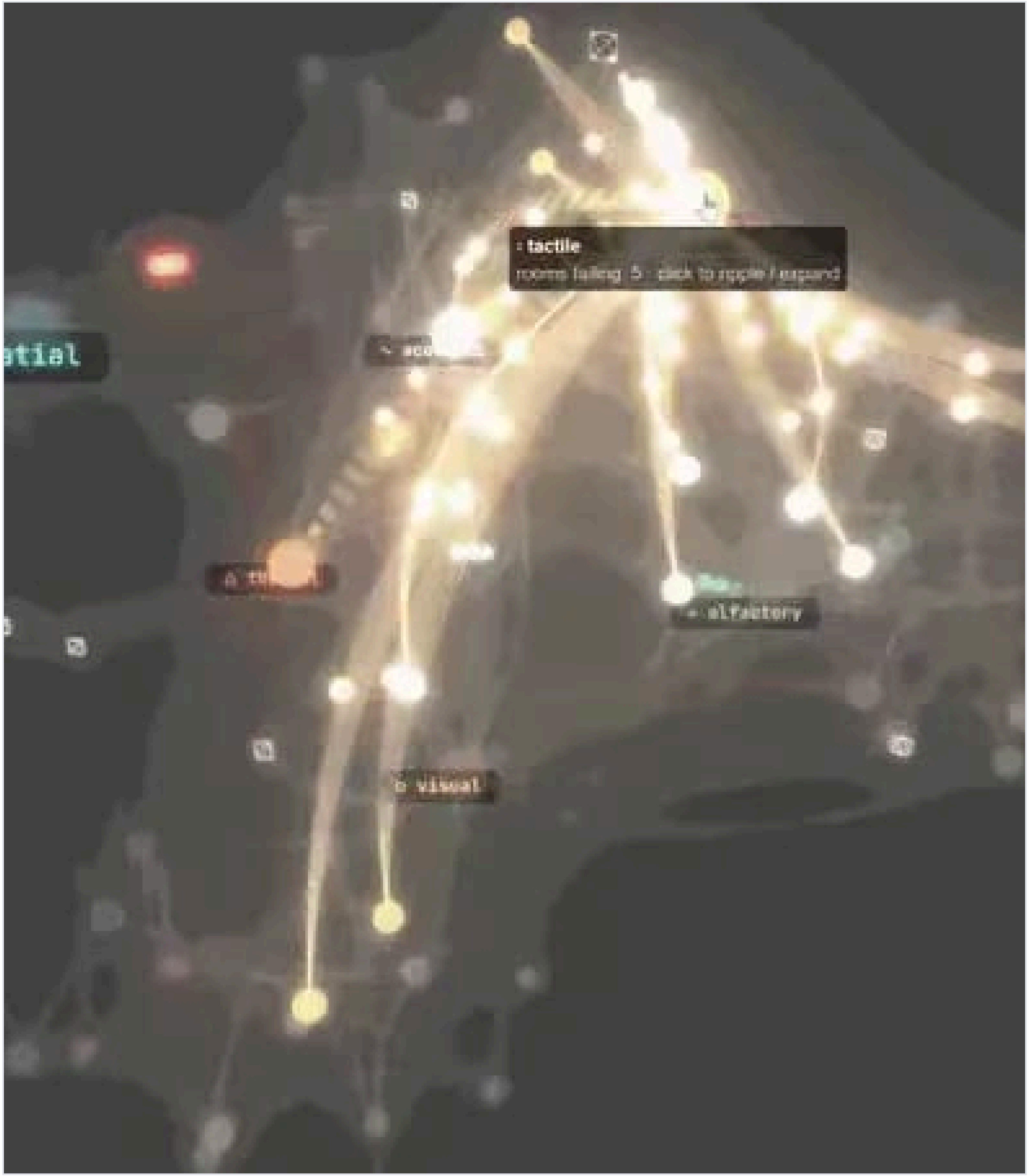
In architecture we model everything: structure, cost, energy, code compliance. Layer after layer that makes a building accountable before it is built. Sensi adds the one layer we never formalized: how a space will actually feel. Not emotion, but the full sensory experience of standing in a room, scored across six senses: thermal, visual, acoustic, spatial, olfactory, tactile. Not in the abstract, but for a specific person.

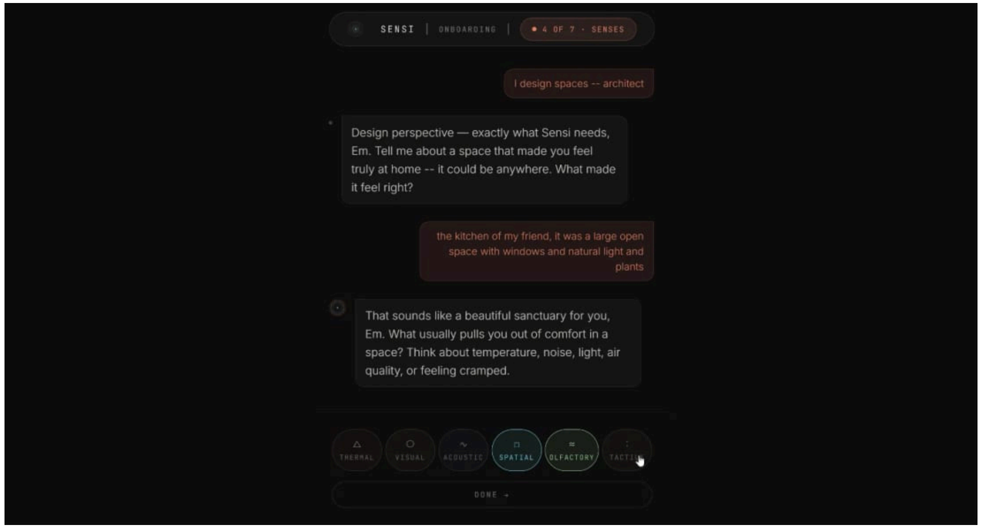
YEAR
2026

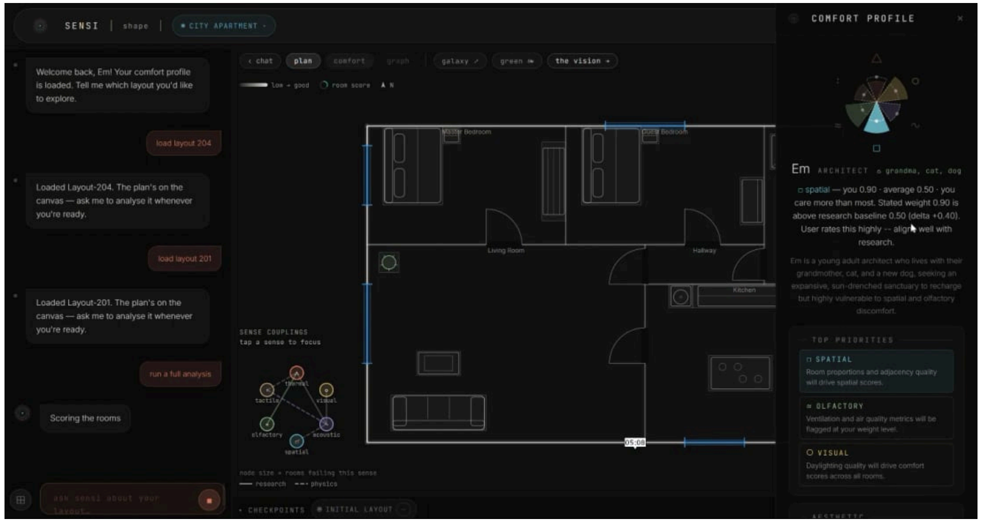
MODULE
AIA Studio · MaCAD, IAAC

TEAM
Charles Abi Chahine, Emilie El Chidiac, Lakzhmy Mari Zaro, María Sánchez Domínguez

TOOLS
Python · LLM agents · Vision models · Web app

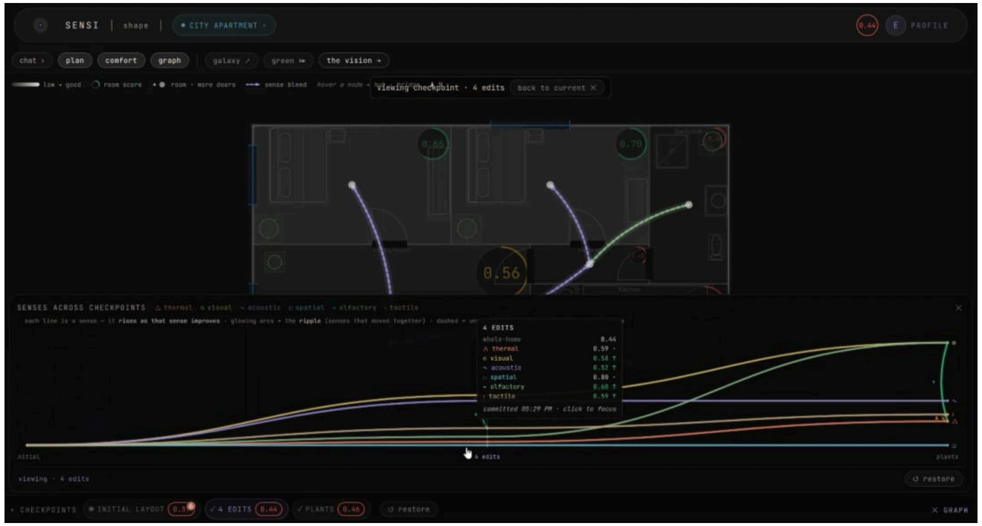


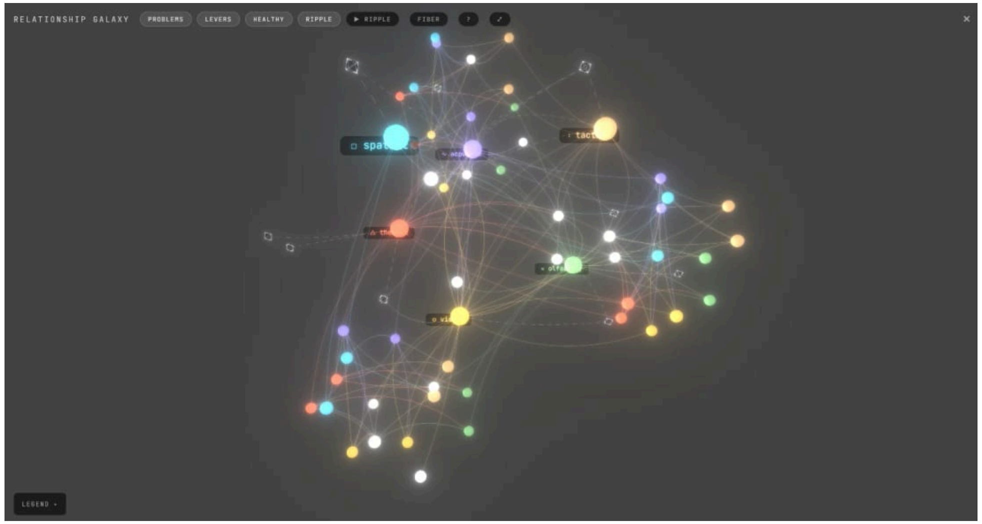




The persona, visualized as a petal rose: your comfort priorities, made explicit.

Reading the plan into rooms, uses, and adjacencies.





Scores respond to every edit, the ripple propagating by fixed rules.

The galaxy view: rooms, senses, and levers as one connected system.

lEgoarCh.

IAAC EXHIBITION

From a one-sentence prompt to a priced, buildable LEGO set: generative AI proposes the form, deterministic computation proves it stands.

You type a building, and a minute later a real LEGO set is sitting on a shelf: rendered, modelled, brick-built, priced, and catalog-legal. Most "AI makes LEGO" demos stop at a gorgeous render, but a render is a promise, not a product. lEgoarCh builds the downstream half: the machinery that forces the dream to obey real bricks, real colours, and real gravity.

YEAR

2026

MODULE

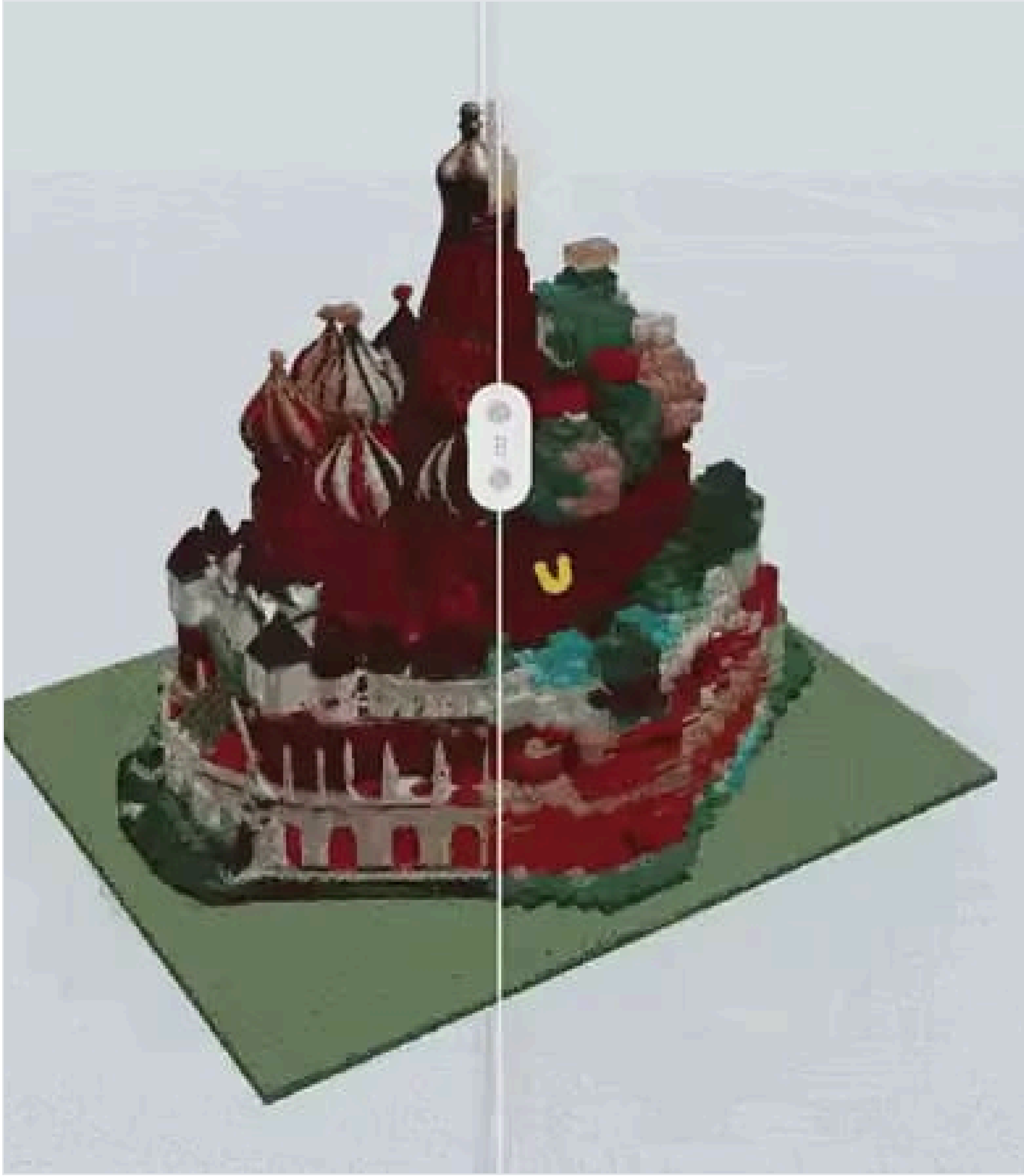
Generative AI Seminar · MaCAD, IAAC

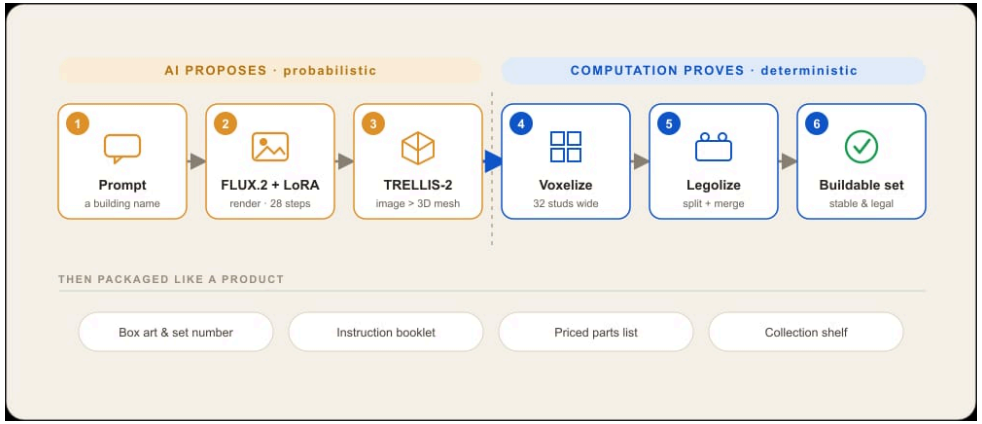
TEAM

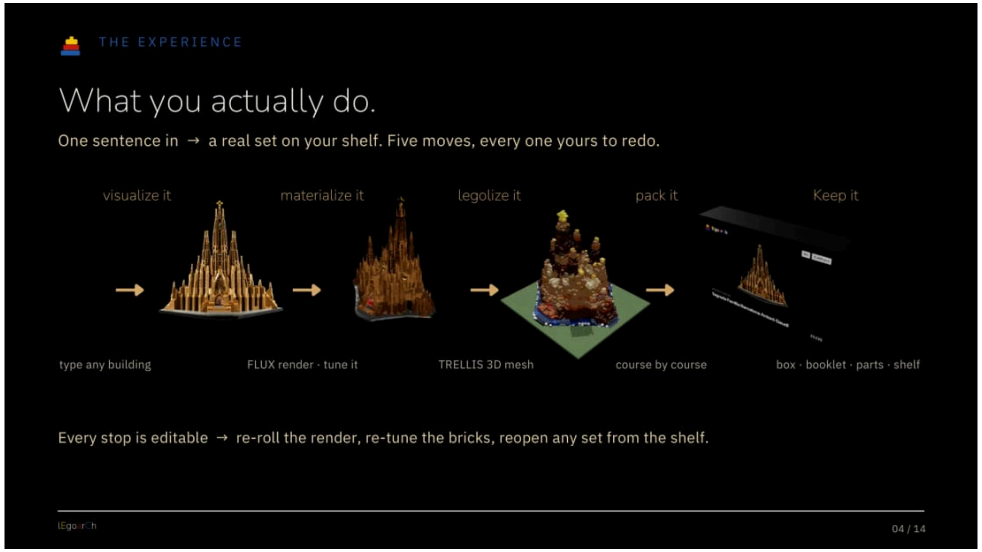
Charles Abi Chahine, Emilie El Chidiac

TOOLS

FLUX.2 + LoRA · TRELLIS-2 · ComfyUI · Python







The whole pipeline on one line: three probabilistic boxes, then three deterministic ones.

The five-move experience: visualize, materialize, legolize, pack, keep.

Same prompt, same seed, only the LoRA strength changes. 1.0 wins.

FLUX.2 + the legoarch LoRA rendering the Sagrada Família as an official-set photo.

Analyzing Narkomfin Through Its Graph.

The Narkomfin Building rebuilt as a spatial graph: centrality and community detection recover Ginzburg's vertical living cells from topology alone, and a GraphSAGE classifier reads room types the plan cannot show.

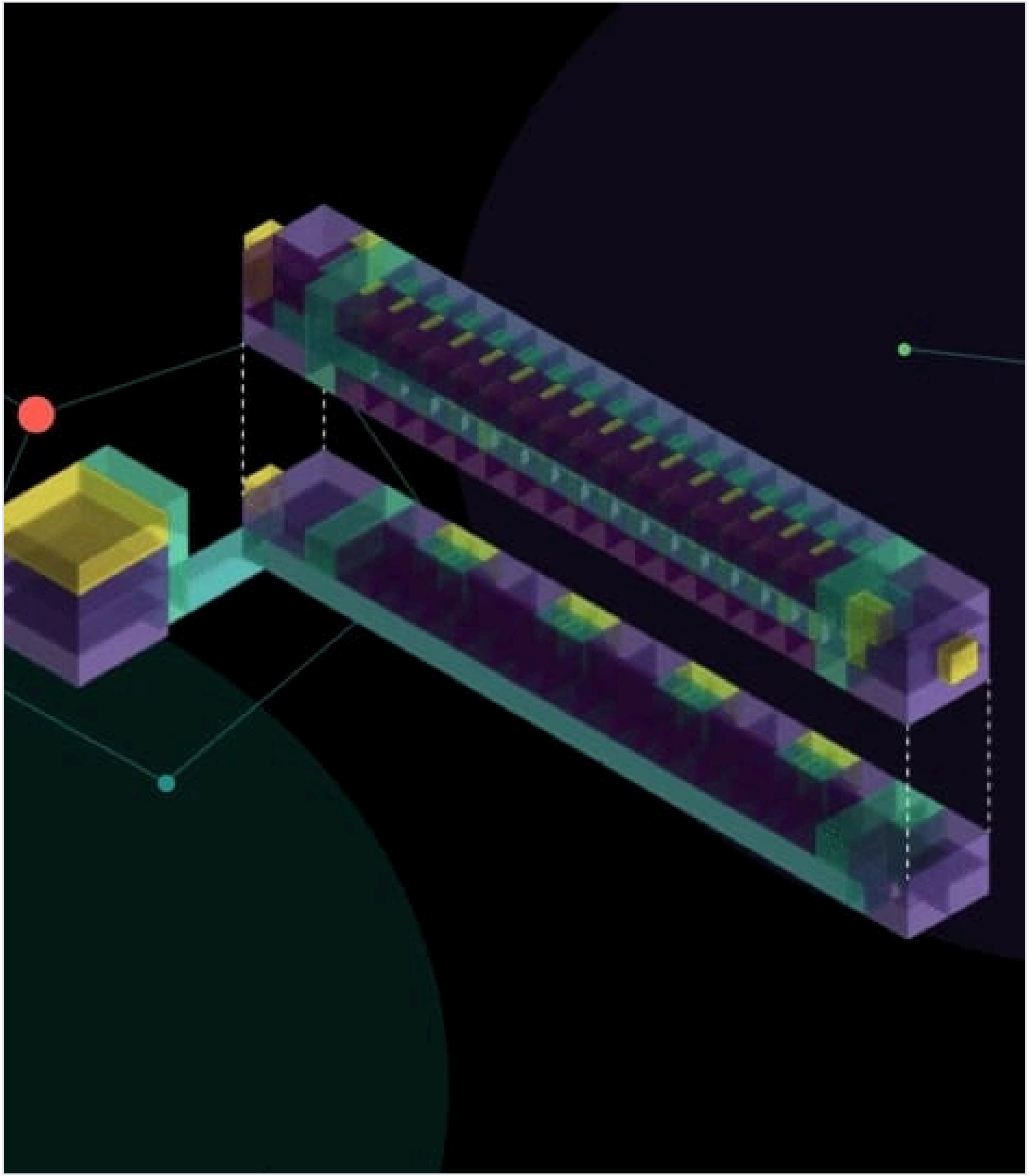
The Narkomfin Building (Moscow, 1930) is reconstructed as a spatial graph to analyse what floor plans cannot show. Grid sampling and stair stitching connect both apartment types into weighted networks; closeness and betweenness centrality expose the corridor as the building's sole topological spine. Community detection recovers Ginzburg's intended vertical living cells from topology alone, and a GraphSAGE room-type classifier scores 68% on the communal Type K layout: confirming the building deliberately breaks domestic spatial conventions.

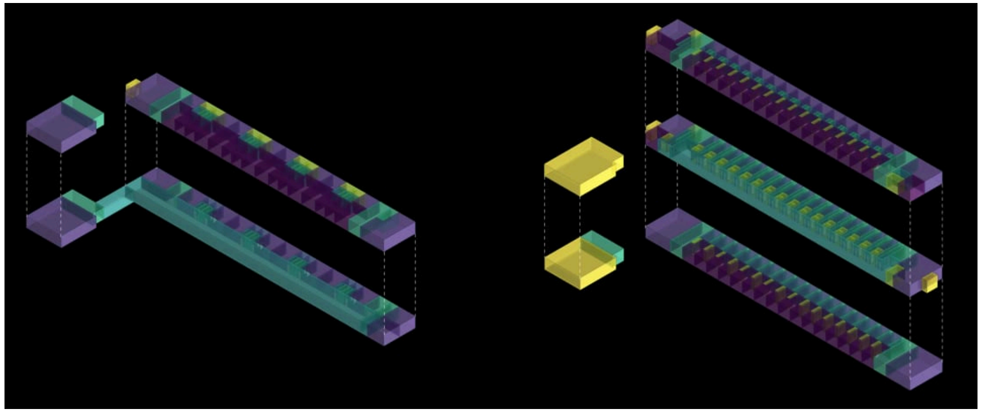
YEAR
2026

MODULE
Graph Machine Learning · MaCAD, IAAC

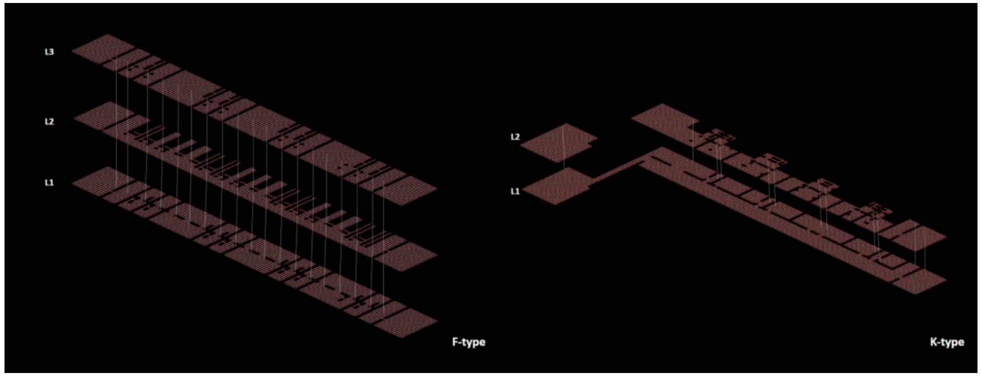
TEAM
Charles Abi Chahine, Emilie El Chidiac, Lakzhmy Mari Zaro, María Sánchez Domínguez

TOOLS
Python · GraphSAGE · Graph ML

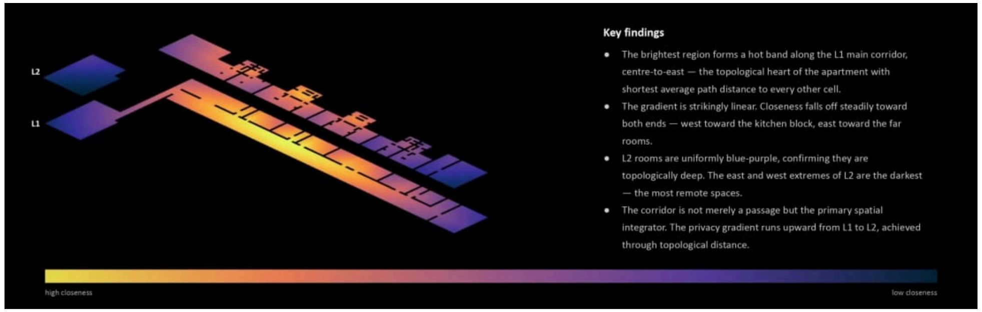




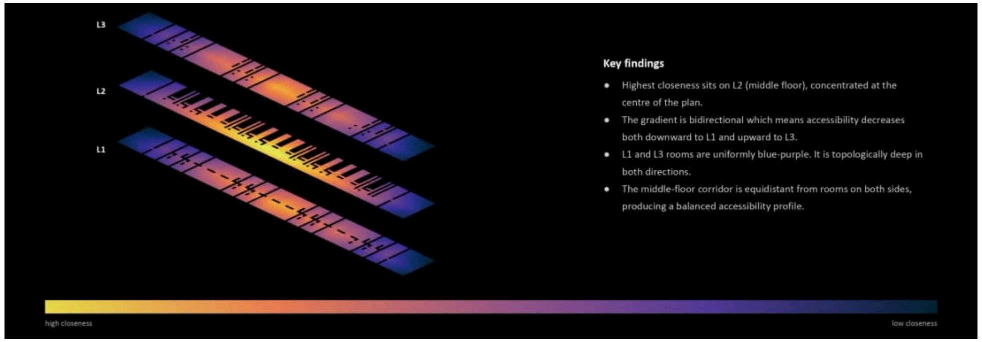
Both apartment types exploded by level. Type K is a two-storey duplex, Type F a three-storey unit for singles, and the corridor sits at a different height in each.



Both types sampled on a 0.5-unit grid and stitched across floors at the stairs. Type K is asymmetric, one dense floor and one fragmented; Type F is symmetric, sparse-dense-sparse.



Type K. The hot band runs the length of the L1 corridor and accessibility falls off in one direction only, upward into the rooms.



Type F. Closeness peaks on the middle floor and drops both downward and upward, a balanced privacy gradient rather than a one-way one.

Encoding Urban Risk.

A machine-learning pipeline that classifies London street segments into low, medium, and high morphological-risk typologies from public spatial data alone, then tests whether the reading survives a move to another city.

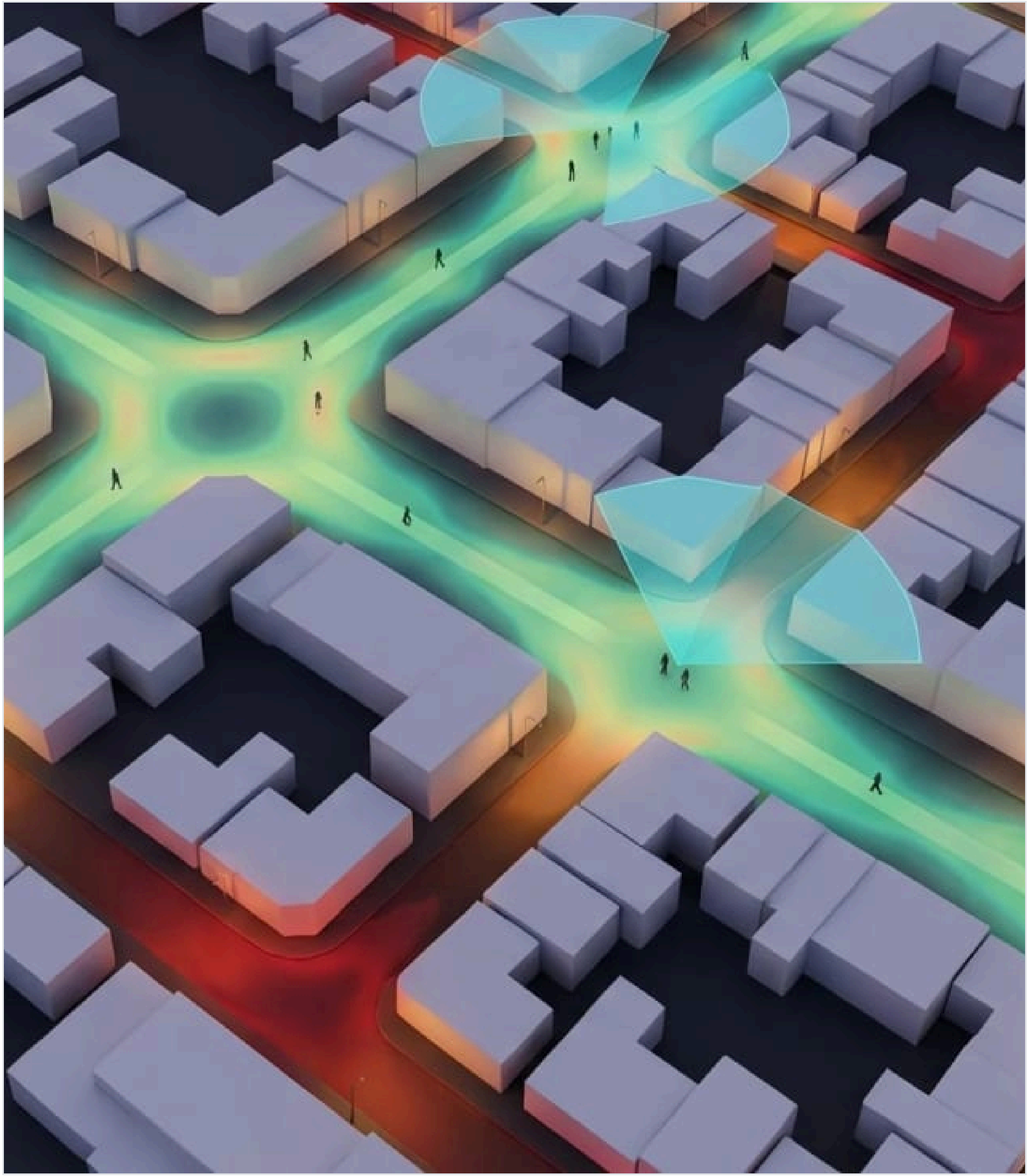
Can the physical layout of a street, measurable from public map data, predict how risky it feels? We framed it as a three-class problem: low, medium, or high risk, using seven spatial features drawn from OpenStreetMap and Mapillary per street segment. This is the full arc of the project, not just the results, but the wrong turn that made them honest.

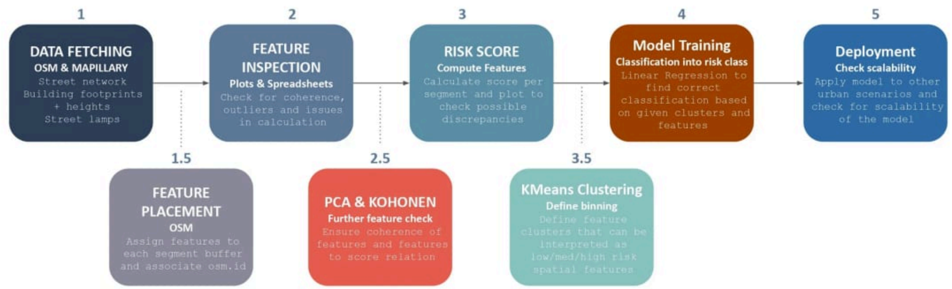
YEAR
2026

MODULE
Data Encoding · MaCAD, IAAC

TEAM
Charles Abi Chahine, Emilie El Chidiac, Lakzhmy Mari Zaro, María Sánchez Domínguez

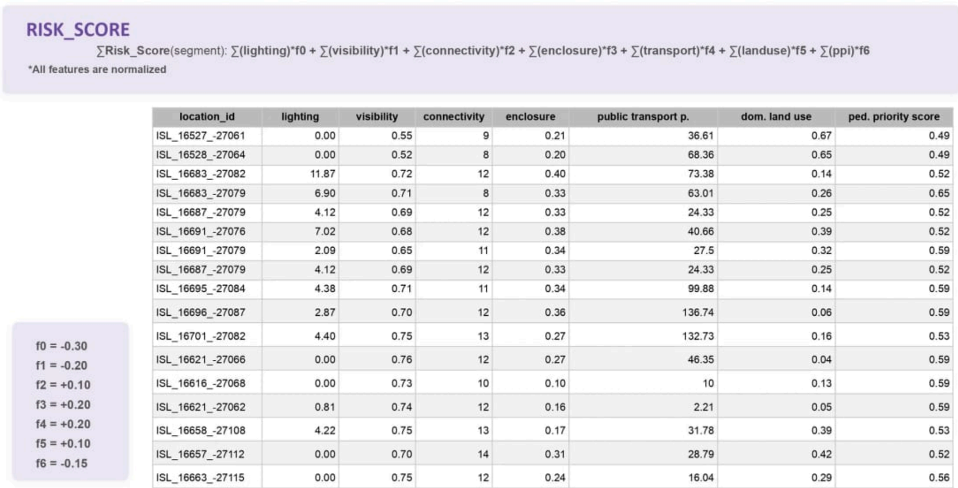
TOOLS
Python · scikit-learn · OpenStreetMap · Mapillary · SHAP





Seven features, each anchored in a lineage of urban-safety theory.

The wall: every model flat against the diagonal. Crime does not reduce to spatial features.



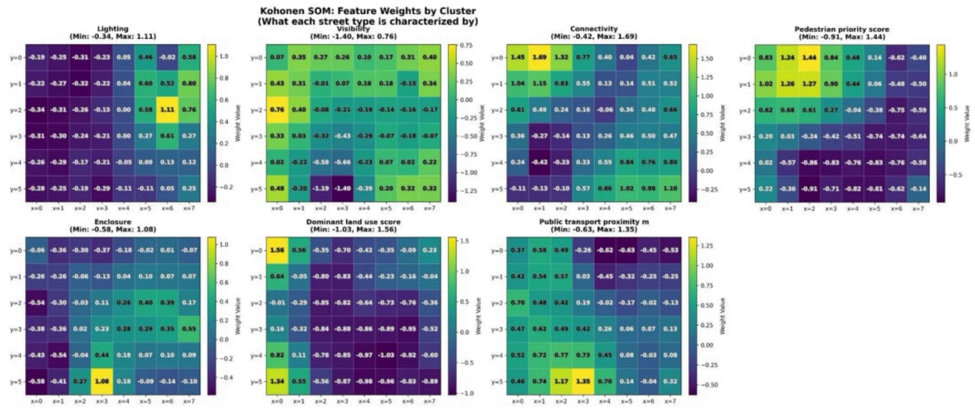
RISK_SCORE

∑Risk_Score(segment): ∑(lighting)*f0 + ∑(visibility)*f1 + ∑(connectivity)*f2 + ∑(enclosure)*f3 + ∑(transport)*f4 + ∑(landuse)*f5 + ∑(ppi)*f6

*All features are normalized

f0 = -0.30
f1 = -0.20
f2 = +0.10
f3 = +0.20
f4 = +0.20
f5 = +0.10
f6 = -0.15

location_id	lighting	visibility	connectivity	enclosure	public transport p.	dom. land use	ped. priority score
ISL_16527_-27061	0.00	0.55	9	0.21	36.61	0.67	0.49
ISL_16528_-27064	0.00	0.52	8	0.20	68.36	0.65	0.49
ISL_16683_-27082	11.87	0.72	12	0.40	73.38	0.14	0.52
ISL_16683_-27079	6.90	0.71	8	0.33	63.01	0.26	0.65
ISL_16687_-27079	4.12	0.69	12	0.33	24.33	0.25	0.52
ISL_16691_-27076	7.02	0.68	12	0.38	40.66	0.39	0.52
ISL_16691_-27079	2.09	0.65	11	0.34	27.5	0.32	0.59
ISL_16687_-27079	4.12	0.69	12	0.33	24.33	0.25	0.52
ISL_16695_-27084	4.38	0.71	11	0.34	99.88	0.14	0.59
ISL_16696_-27087	2.87	0.70	12	0.36	136.74	0.06	0.59
ISL_16701_-27082	4.40	0.75	13	0.27	132.73	0.16	0.53
ISL_16621_-27066	0.00	0.76	12	0.27	46.35	0.04	0.59
ISL_16616_-27068	0.00	0.73	10	0.10	10	0.13	0.59
ISL_16621_-27062	0.81	0.74	12	0.16	2.21	0.05	0.59
ISL_16658_-27108	4.22	0.75	13	0.17	31.78	0.39	0.53
ISL_16657_-27112	0.00	0.70	14	0.31	28.79	0.42	0.52
ISL_16663_-27115	0.00	0.75	12	0.24	16.04	0.29	0.56



The eight-step pipeline: from OSM features to a deployable classifier.

Redundancy check: the one strong correlation is visibility vs. enclosure at −0.75.

Breathing Mass.

A hyperbuilding for Santiago conceived as a vertical lung: a tower that captures, cleans, and redistributes the city's polluted air through a breathing core, resolved as a topology-optimized structure and a performance-driven facade.

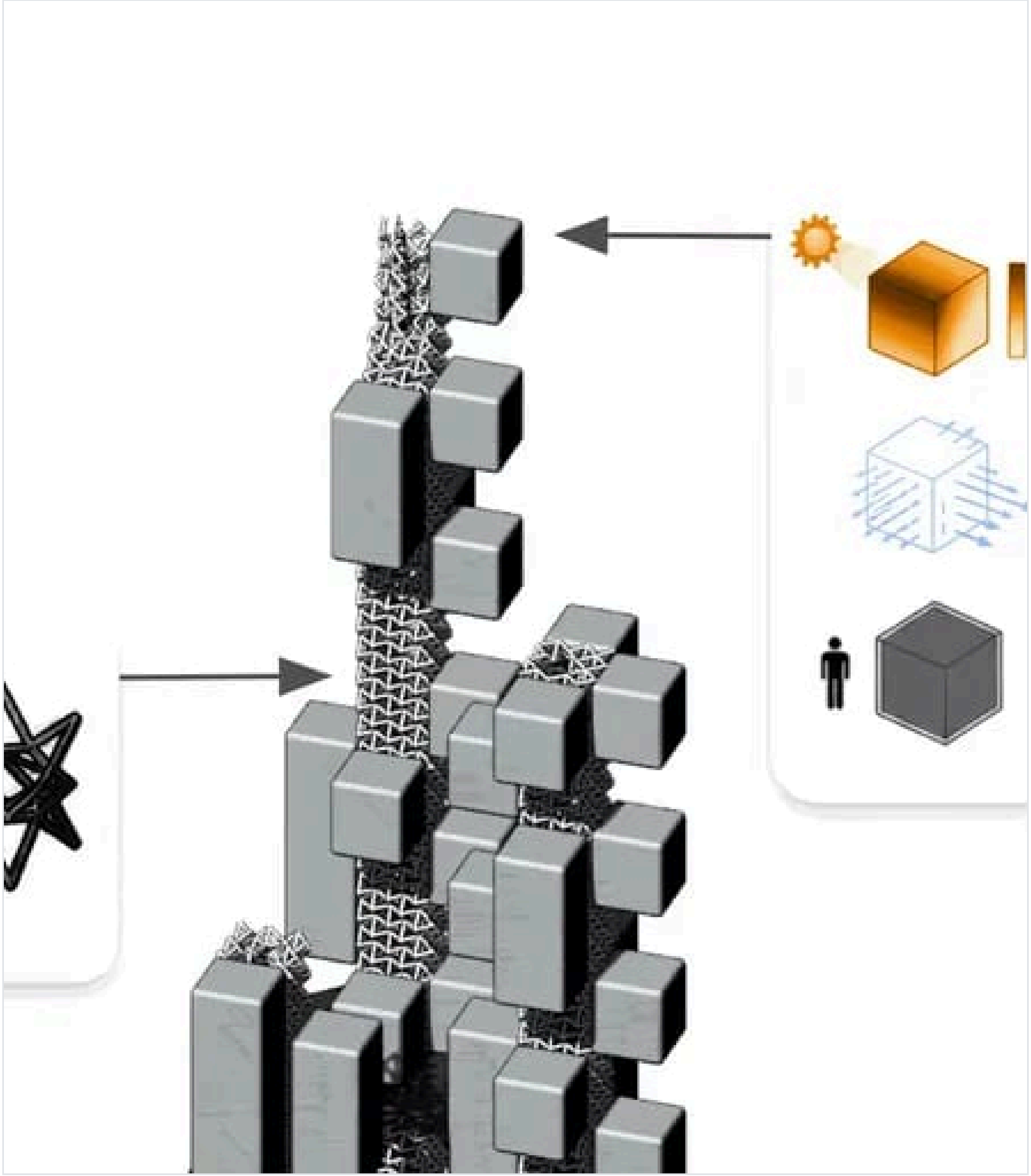
Breathing Mass is a vertical ecosystem in Santiago where architecture, wind, and energy converge. The Hyper Lung captures, cleans, and redistributes polluted air through a breathing core, turning the tower into living infrastructure that breathes with the city. Within a larger hyperbuilding studio, our team owned its structure and facade: translating the lung analogy into a self-braced skeleton and an adaptive skin, every form justified by data and published to the program and data teams through Speckle.

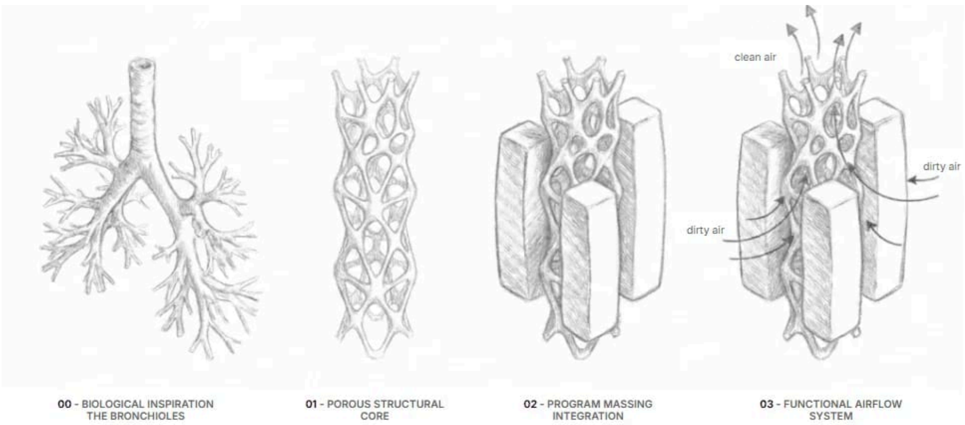
YEAR
2026

MODULE
BIMSC Studio · MaCAD, IAAC

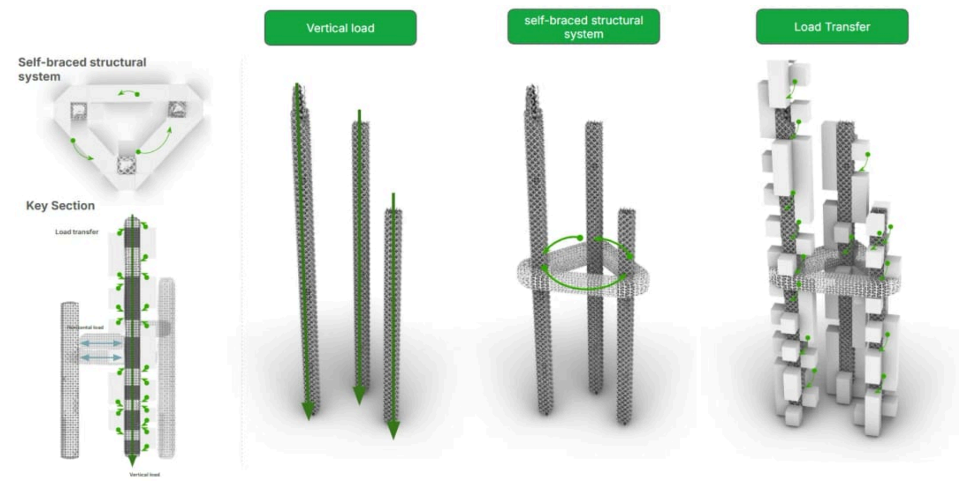
TEAM
Charles Abi Chahine, Ramy Ayoub, Hani Karime

TOOLS
Rhino · Grasshopper · Alpaca · Speckle

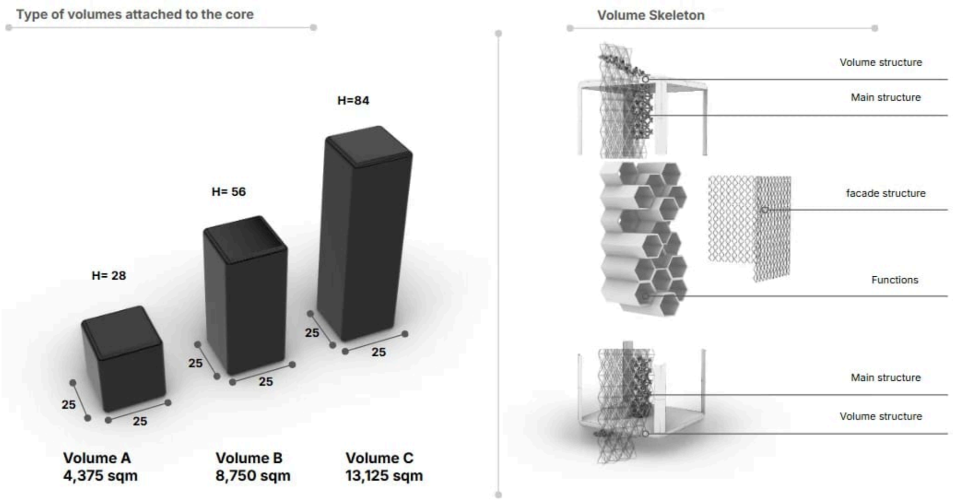




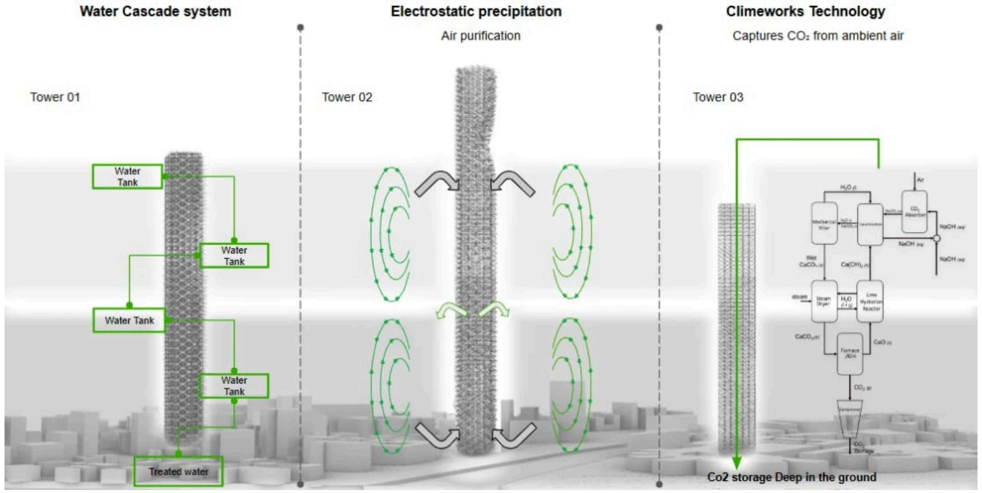
From bronchi to a porous structural core: the lung analogy driving the spine.



Self-braced structural system: load transfer from plugged volumes through the triangulated cores.



Volume scattering: candidate masses evaluated by radiation, wind, and density.



The environmental machine: water cascade, electrostatic precipitation, and CO_2 capture.

The Huddle.

IAAC EXHIBITION

A wind-adaptive research & education hub with expedition basecamp in Punta Arenas, Chile: discrete modules aggregated against a perpetually windy subpolar climate.

Punta Arenas sits at the southern tip of Chile in a cold, perpetually windy subpolar climate. The Huddle is a research and education hub with an expedition basecamp for 1,000 users, designed the way penguins survive the same latitudes: compact forms, packed close, shielding one another from the wind.

YEAR

2025/26

MODULE

ACESD Studio · MaCAD, IAAC

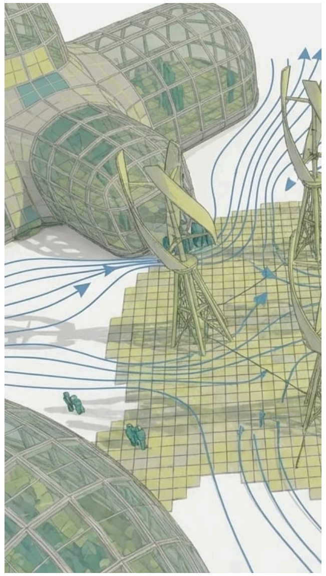
TEAM

Charles Abi Chahine, Emilie El Chidiac, María Sánchez Domínguez, Lakzhmy Mari Zaro

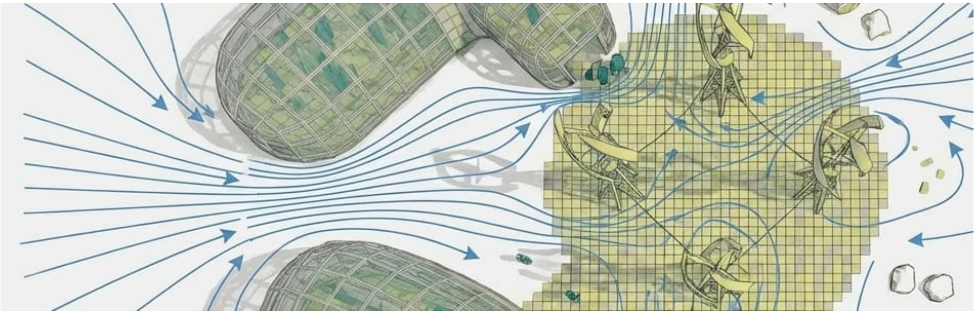
TOOLS

Rhino · Grasshopper · Wasp · Kangaroo · Speckle

Wind analysis over the aggregation.



Flow around the massing.



Iterating the aggregation against the wind field.

The kit of parts, aggregated.

The Luminous Stratum.

A “volume of sedimented light” inserted into Cairo’s historic fabric: an independent lattice that hovers within a market void, filtering the harsh sun without ever touching the historic walls.

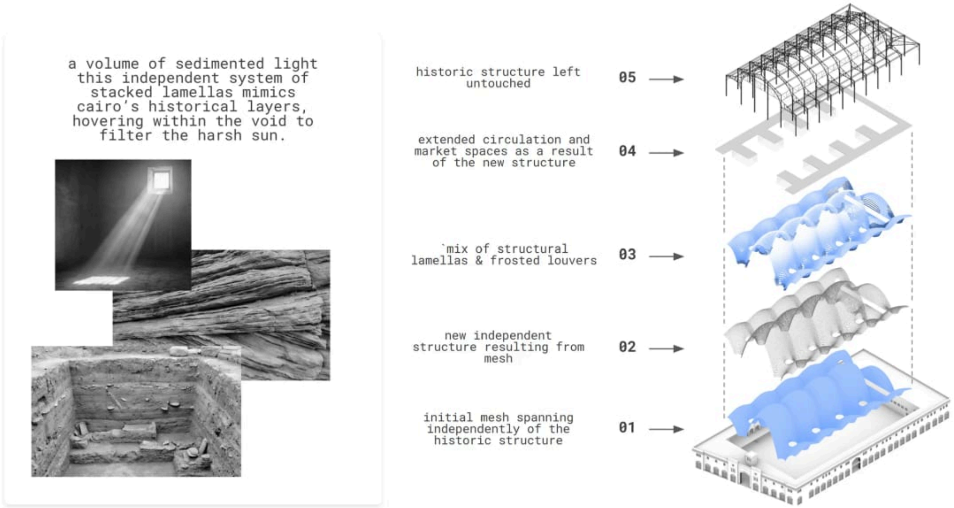
In the dense historic fabric of Cairo, light is both a blessing and a burden. The Luminous Stratum proposes a new architectural language that negotiates this relationship: a “volume of sedimented light” that mimics the city’s stratification while filtering the harsh sun. Designed for the Complex Forming seminar, it is an independent system of stacked lamellas that hovers within the void of the Bab Al-Luk historic market, leaving the original structure untouched.

YEAR
2025

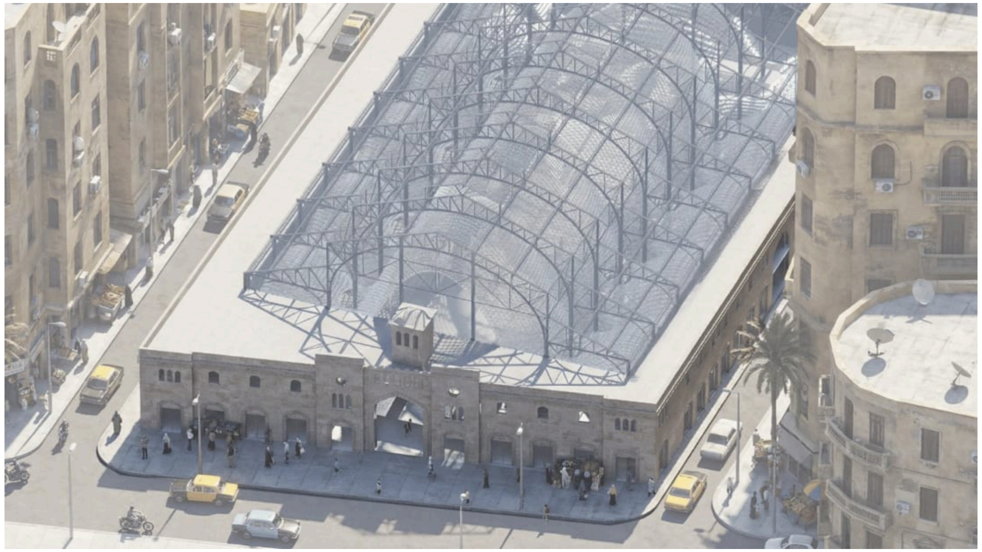
MODULE
Complex Forming · MaCAD, IAAC

TEAM
Charles Abi Chahine

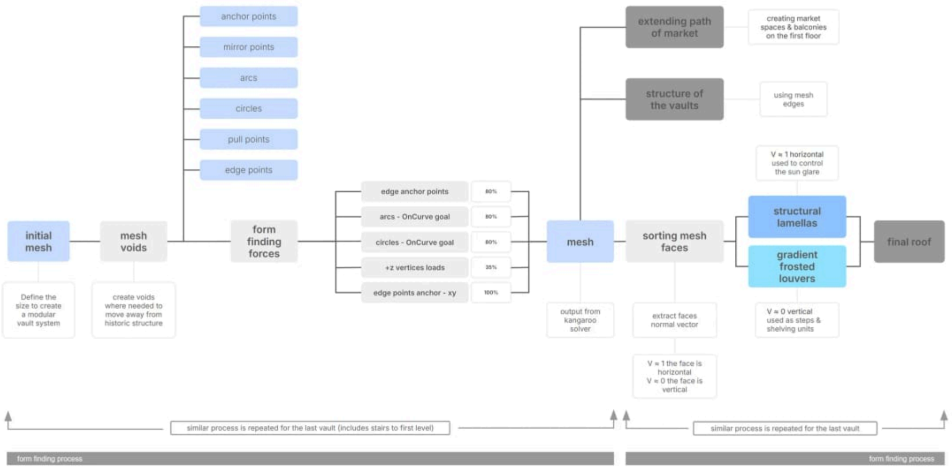
TOOLS
Rhino · Grasshopper · Kangaroo



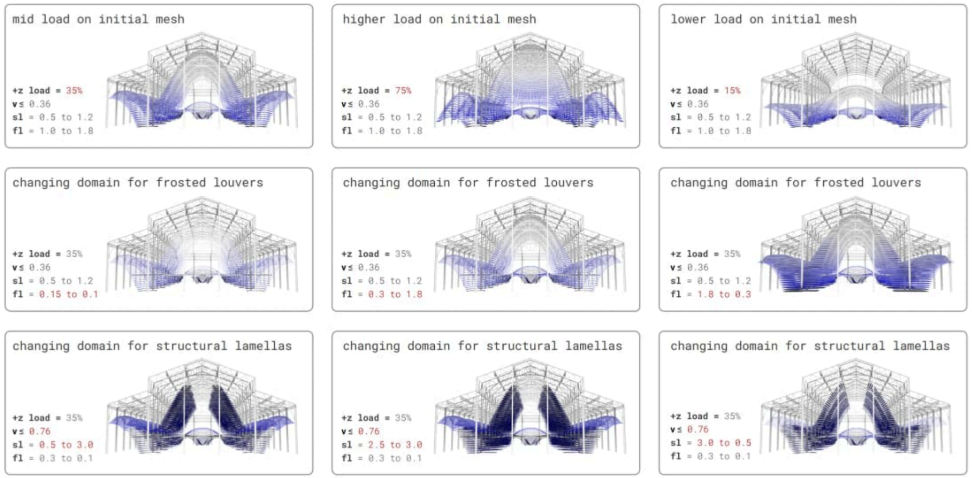
A volume of sedimented light: an independent system layered above the untouched historic market.



The lattice hovering within the historic market hall.



The computational logic: anchor points, form-finding forces, and the sorting of mesh faces.

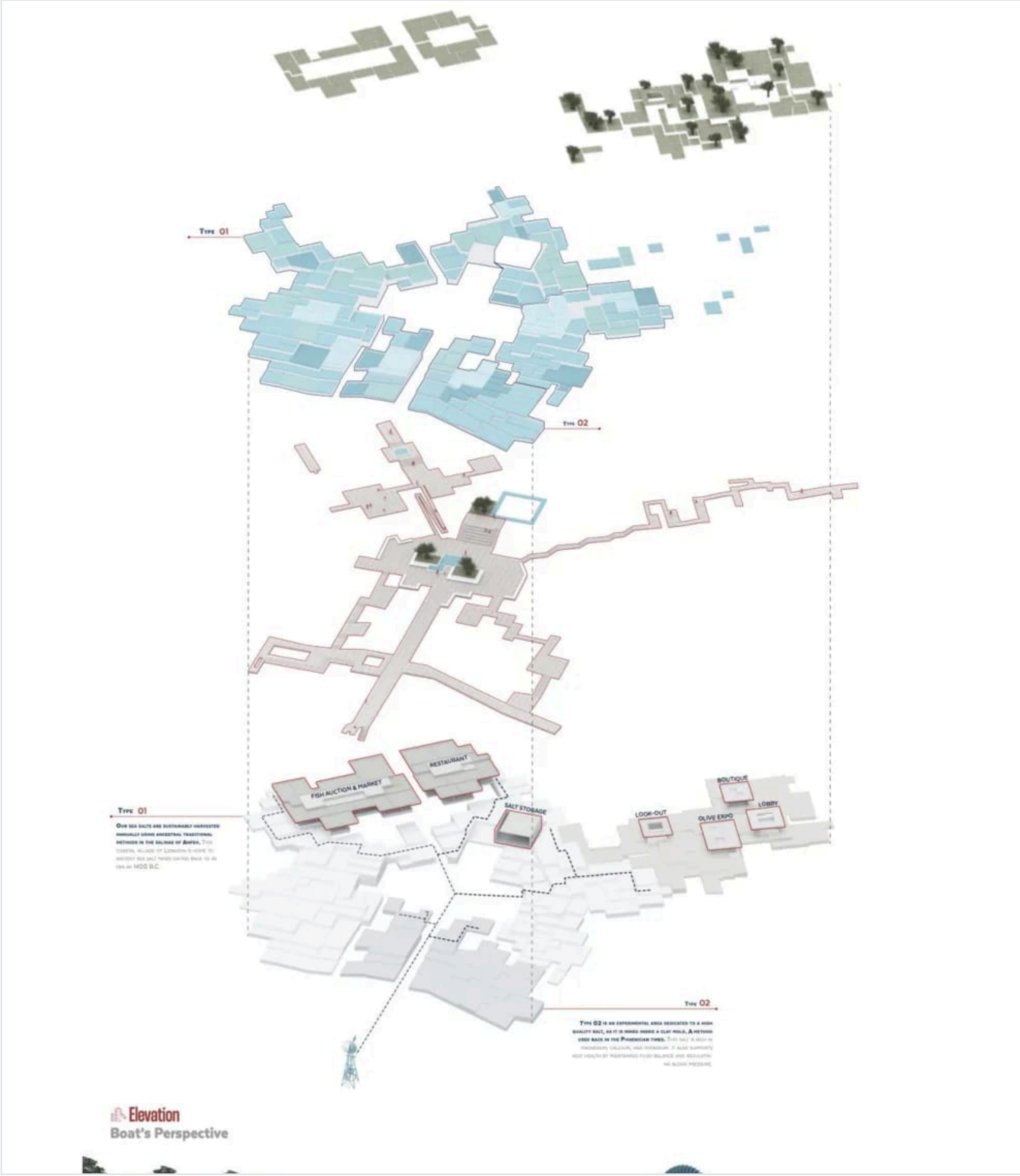


Iterating vertical loads and louver/lamella domains to balance shading against structure.

Anfeh.

My bachelor final-year thesis at LAU: reimagining the coastal town of Anfeh in North Lebanon through its landscape of salt ponds, fishing, and olives.

Anfeh is my final-year thesis at the Lebanese American University (Spring 2023), advised by Issam Barhouch and Omar Harb. It reimagines the rural coastal town of Anfeh in North Lebanon, working from its landscape of salt ponds, fishing, and olives toward an urban re-imagination of the town.

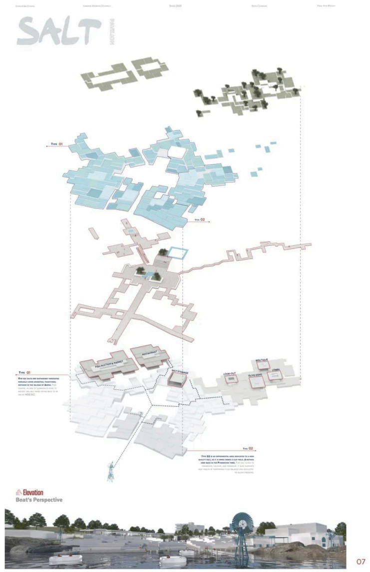


YEAR
2023

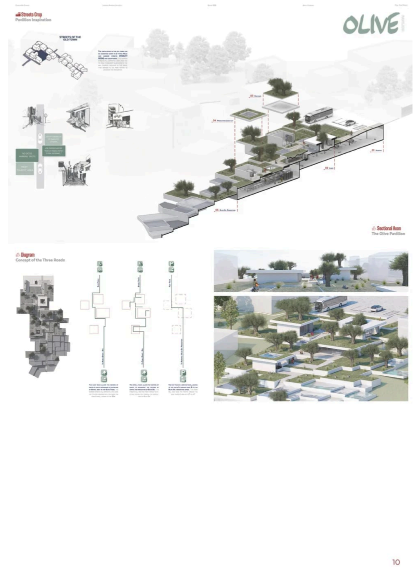
MODULE
B.Arch Final Year Project: LAU

TEAM
Charles Abi Chahine

TOOLS
Architecture · Urban design



From the Anfeh final-year portfolio.



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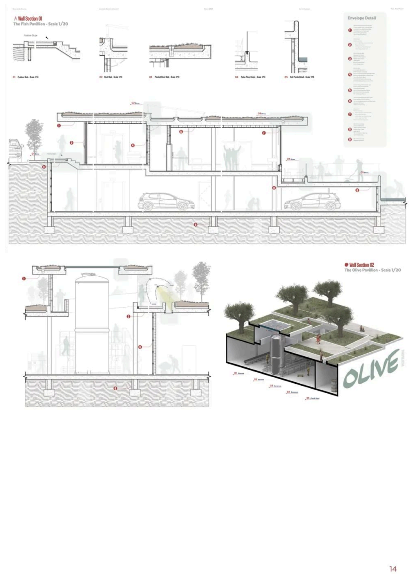

From the Anfeh final-year portfolio.

THE REST OF IT

Eighteen projects, four belts.

This booklet is a selection. Every project, with its full gallery and write-up, is on the site.

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